Natural Language as a Whole: Detecting Generated Texts in Maltese via Clustering and Topological Analysis

Matvey Safonov

Supervisor: Prof. V. A. Gromov

National Research University Higher School of Economics

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Abstract

This work was carried out within the *Spot the bot* project of V. A. Gromov's laboratory and addresses the task of distinguishing texts written by humans from texts generated by a neural language model, for the **Maltese language** – a low-resource language of the Semitic family. A corpus of 36,452 human texts was assembled, a word-level LSTM generator was trained, and for both corpora clouds of n-gram vectors based on SVD embeddings were obtained. The clouds were studied with Wishart clustering and persistent homology. We show that the cloud of bot texts differs from the human one in a statistically significant way: it breaks into a larger number of clusters, has a higher noise fraction, is worse separated (Davies–Bouldin criterion), and is topologically more complex (more H_1 cycles). The differences are confirmed by a bootstrap estimate ($B = 40$, Mann–Whitney test, $p < 10^{-5}$, Cohen's effect size 1.2–2.3).

Keywords: natural language processing, Maltese language, SVD, latent semantic mapping, LSTM, Wishart clustering, persistent homology, generated-text detection.

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1 Introduction

The hypothesis that natural language is a *self-organized critical system* has been developed in a series of works by V. A. Gromov's laboratory [1, 2]. In this paradigm a text is regarded not as a linear sequence but as a holistic object possessing geometric and topological structure in the space of semantic vectors. If the structure of human language is organized in a particular way, then a text produced by an artificial language model should differ from a human one precisely at the level of this structure, even when locally (at the level of individual n-grams) it looks plausible.

The *Spot the bot* project sets the task of testing this hypothesis: to learn to **tell a generated text from a human one** by the geometry of the cloud of its n-grams. The project methodology is applied to different languages; the present work carries out the full cycle for the **Maltese language** (`mt`).

Maltese is an interesting object of study: it is the only Semitic language with a Latin script and official status in the EU, while at the same time being low-resource – ready-made neural models and large corpora for it are scarce. This makes the task meaningful from the engineering side as well.

Aim of the work – to run the full *Spot the bot* pipeline for Maltese and to establish whether the human and generated corpora are distinguishable by methods of clustering and topological analysis.

Objectives:

1. collect and preprocess a corpus of Maltese texts;
2. build a dictionary of SVD embeddings;
3. train a neural generator and obtain the bot corpus;
4. form clouds of n-gram vectors for both corpora;
5. cluster the clouds with the Wishart algorithm and compare quality metrics;
6. study the clouds by means of persistent homology;
7. assess the statistical significance of the differences.

2 Problem statement

Let a corpus of human texts $\mathcal{H}$ and a corpus of generated texts $\mathcal{B}$ of comparable size be given. Each word is mapped to a vector of fixed dimension m; n consecutive words give an n-gram vector of dimension $n \cdot m$ (concatenation). The set of all n-grams of a corpus forms a *point cloud* in $\mathbb{R}^{nm}$.

Formally, we test the null hypothesis

$$H_0: \quad \text{the clouds } \mathcal{H} \text{ and } \mathcal{B} \text{ are statistically indistinguishable by structural characteristics,}$$

against the alternative H_1 of their difference. As structural characteristics we take: the number and sizes of clusters and the cluster-validity metrics (internal validity measures, Ch. 23 [5]), as well as topological invariants – Betti numbers and characteristics of persistence diagrams.

3 Theoretical background

3.1 SVD embeddings and latent semantic mapping

The main type of embeddings in the project methodology is embeddings based on the singular value decomposition, going back to Bellegarda's latent semantic mapping [3]. From the cleaned corpus a weighted term–document matrix $A \in \mathbb{R}^{V \times D}$ is built (V – the number of unique words, D – the number of texts) with TF-IDF weights. The truncated singular value decomposition of rank k

$$A \approx U_k \Sigma_k V_k^{\top}, \qquad U_k \in \mathbb{R}^{V \times k}, \;\; \Sigma_k = \mathrm{diag}(\sigma_1, \dots, \sigma_k),$$

gives a vector representation of a word: the i-th row of $U_k \Sigma_k$ corresponds to the k-dimensional vector of the i-th word. An essential property of the method: to obtain an embedding of smaller dimension $d < k$ no repeated decomposition is needed – it suffices to take the first d components of the vector. The decomposition is therefore carried out once with a large k (here $k = 1024$).

3.2 Wishart clustering

The Wishart algorithm [4] is a density-based clustering method (*mode analysis*) that isolates clusters as regions of high density separated by regions of low density. The density at a point x_i is estimated from the distance $r_k(i)$ to its k-th nearest neighbour:

$$p(x_i) = \frac{k}{n \cdot V_d \cdot r_k(i)^{\,d}}, \qquad V_d = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)},$$

where n – the number of points, d – the dimension, V_d – the volume of the unit d-dimensional ball. Points are processed in order of decreasing density; for the current point the already-processed clusters it is connected to are determined via the k-nearest-neighbour graph:

- no connections – a new cluster is formed;
- one cluster – the point joins it (if it is not "completed");
- several clusters – the *significant* clusters (with a density gap $\geq h$) are frozen and the point goes to noise; the insignificant ones are merged.

The parameters of the algorithm are k (the number of neighbours for the density estimate) and h (the significance threshold). The quality of the resulting clustering is assessed by internal validity measures (Ch. 23 [5]): silhouette, Davies–Bouldin index, Calinski–Harabasz index.

3.3 Persistent homology

Persistent homology [6, 7] is a tool of topological data analysis. From a point cloud a filtration of Vietoris–Rips complexes is built: as the parameter ε grows, points that are within distance $\leq \varepsilon$ are joined by edges and simplices are formed. During the filtration topological features (H_0 – connected components, H_1 – one-dimensional cycles) are "born" and "die". The persistence diagram records pairs (birth, death); long-lived features are considered significant, short-lived ones – noise. Numerical characteristics: Betti numbers b_0, b_1, total and maximal persistence, persistence entropy

$$E = -\sum_i \frac{\ell_i}{L} \log \frac{\ell_i}{L}, \qquad \ell_i \; - \; \text{the lifetime of the } i\text{-th feature}, \;\; L = \sum_i \ell_i.$$

4 Corpus collection and preprocessing

4.1 Sources

The corpus of Maltese texts was assembled from open sources (see Table 1): the academic corpus of the University of Malta (MLRS / Korpus Malti), the UM research-output repository, the Maltese Wikipedia, blogs and press, and non-fiction texts. In total over 110 thousand raw text files were collected; to balance the sources, blogs and press were capped at 8,000 texts each.

Table 1: Composition of the human corpus after source balancing

Source	Texts
UM research-output repository (`umlib_oar`)	11,280
Blogs (`blogs`)	8,000
Press (`press_mt`)	8,000
Wikipedia (`wiki`)	7,041
Non-fiction (`nonfiction`)	2,113
Theses (`theses`)	18
Total	**36,452**

4.2 Preprocessing

Preprocessing reduces word forms to their base form (lemmatization) and replaces function parts of speech with special tokens: pronouns – with `PRON1`, numerals – with `ORDINAL1`, proper names – with `PERSON1`; punctuation and symbols are discarded. The replacements rely on the part of speech, so the lemmatizer needs POS tagging.

Three lemmatization variants. Since Maltese is low-resource and no reliable ready-made lemmatizer exists for it, three variants were implemented and compared:

- `L1` – the neural lemmatizer `Stanza` (the `tokenize` / `pos` / `lemma` pipeline). POS tagging is reliable, but neural lemmas for a low-resource language are often inaccurate and not backed lexically by anything.
- `L3` – a purely rule-based preprocessor without NLP libraries (strippers of Maltese affixes: hyphenated articles `il-`, `it-`, …, apostrophe proclitics `b'`, `f'`, …, possessive and object suffixes), close in spirit to Subkhangulov's lemmatizer for Tatar. It is "strict by the methodology" but does not yield parts of speech, so the function-word replacements are performed only heuristically (by lists and case) rather than by the actual POS.
- `L2` – a combined one: the POS tag and the base lemma are taken from `Stanza`, after which our own *lexicon* lemmatizer is applied, based on the Maltese morphological database Ġabra (1,303,378 word forms).

How the L2 lemmatizer works. For each word form w (if its part of speech is not a function one and is not discarded) the lemma is determined by a cascade of lookups in the Ġabra table; the first step that fires gives the result:

1. exact lookup of w (lower-cased) in Ġabra;
2. lookup of the form with diacritics removed (`ċġħż`→`cghz`);

3. lookup after stripping the article/proclitic (`il-`, `b'`, ...), including without diacritics;
4. fallback to the lemma proposed by `Stanza`;
5. if nothing is found – the original word form itself.

Thus L2 combines the strengths of L1 and L3: a reliable POS tag for the function-word replacements – and an accurate, lexically verified lemma instead of an unreliable neural one. The distribution of the sources of the final lemmas across the corpus: Ġabra – $\approx 4.1\,\%$ of tokens (form correction), the `Stanza` neural lemma – $< 0.1\,\%$, the remaining $\approx 95.9\,\%$ coincide with the original form (the word is already in its base form or is absent from Ġabra). The low share of actually changed tokens reflects the analyticity of the corpus and the coverage limits of the lexicon for a low-resource language.

Choice of variant. `L2` was chosen as the working variant: it yields the most accurate genuinely Maltese lemmas while preserving the POS replacements required by the methodology, whereas `L1` is lexically weaker and `L3` loses information about parts of speech. The entire downstream pipeline (SVD dictionary, generator, n-grams, clustering, topology) is built on the `L2` corpus. Result: 36,452 cleaned texts, about 39.3 million word tokens; the output is a single file, one line – one text.

5 Building the embeddings

5.1 SVD dictionary

From the cleaned corpus, a term–document matrix was built using `TfidfVectorizer`. The Maltese alphabet (including the letters "ċ, ġ, ħ, ż") together with digits, hyphen and apostrophe was given as the `token_pattern` – this cuts off punctuation and foreign-language insertions. A sparse matrix of size $93{,}343 \times 36{,}452$ (density 0.0032) was obtained.

A truncated singular value decomposition of rank $k = 1024$ was applied to the matrix. The singular values decrease from $\sigma_1 \approx 39.7$ to $\sigma_{1024} \approx 2.3$. The result is a dictionary $\{\text{word} \mapsto \text{vector} \in \mathbb{R}^{1024}\}$ for 93,343 words.

5.2 word2vec

For comparison, word2vec embeddings were also built (CBOW and Skip-gram, dimension 100, window 5) on the same corpus with `gensim` (vocabulary – 107,405 words).

6 Generating the bot corpus

The bot corpus is produced by a neural generator trained on the human corpus. A **word-level LSTM** language model was used: it operates on the same lemma tokens as the SVD dictionary, so the generated texts are directly suitable for building n-grams.

Architecture and training. Model: an embedding layer (300), a two-layer LSTM (hidden size 512, dropout 0.3), a linear layer onto the vocabulary (the 50,000 most frequent words); 44.4 million parameters in total. Training – next-token prediction with truncated BPTT, the Adam optimizer, 8 epochs. The perplexity on the held-out set decreased from 164 (epoch 1) to **98** (epoch 8).

Generation. From the trained model, 36,452 texts were generated – the same number as in the human corpus. The length of each text is drawn from the empirical distribution of human text lengths, which ensures comparability of the corpora by volume. The resulting bot corpus: 36,452 texts, 29.6 million tokens.

7 The n-gram dataset

For each corpus, n-grams are extracted with a sliding window of length n; the vector of each word is taken from the SVD dictionary (the first m components), and the word vectors of the n-gram are concatenated into a vector of dimension $n \cdot m$. If at least one word of the n-gram is absent from the dictionary, the n-gram is skipped.

The clouds were formed for several configurations (n, m); each cloud contains 300,000 n-gram vectors for the human and for the bot. The fraction of n-grams skipped because of a missing word grows with n (for $n = 3$ – about 20 %, for $n = 5$ – over 95 %), which is natural: the probability that *all* words of a long n-gram are in the dictionary falls as n grows.

8 Wishart clustering: experiments

8.1 Parameter sweep

First, the sensitivity of the algorithm to the parameters k and h was studied on the configuration $n = 4,\ m = 8$. The significance parameter h turned out to be practically inert: when varied from 0.2 to 125 the results did not change – in a high-dimensional cloud the density gaps within clusters greatly exceed any reasonable h. The parameter k, on the contrary, strongly affects the granularity of the partition: $k = 7$ gives about 400 clusters, $k = 51$ – about 30. The noise fraction is high for any parameters (75–87 %) – this is a structural property of density-based clustering in a high-dimensional space. For the further experiments $k = 11,\ h = 1$ were fixed.

8.2 Comparison of configurations

Six configurations were compared (embedding type $\times$ parameters n, m), Table 2. In *all* configurations the bot cloud breaks into a *larger* number of clusters than the human one ($\Delta_{\text{cl}} = +13 \ldots + 31$). The word2vec embeddings, truncated to 8 components, cluster degenerately (the silhouette goes into the negative range), which confirms the appropriateness of the SVD embeddings. The best human/bot divergence across the metrics taken together is given by the configuration **SVD,** $n = 4$**,** $m = 16$.

Table 2: Comparison of configurations (Wishart, $k = 11$, $h = 1$, 30,000 n-grams each). H – human, B – bot

Emb.	(n, m)	dim	H cl.	B cl.	H noise	B noise	Δ cl.
SVD	$(3, 8)$	24	321	352	64.2 %	65.1 %	+31
SVD	$(4, 8)$	32	234	247	77.9 %	79.4 %	+13
SVD	$(5, 8)$	40	156	169	84.6 %	83.6 %	+13
SVD	$(4, 16)$	64	273	302	71.2 %	75.1 %	+29
SVD	$(4, 32)$	128	305	322	62.8 %	64.3 %	+17
word2vec	$(4, 8)$	32	17	30	93.5 %	92.6 %	+13

8.3 The final configuration

The full analysis on the configuration SVD, $n = 4$, $m = 16$ (Table 3) shows: the bot cloud has more clusters, more noise, a higher Davies–Bouldin index (worse separated) and a higher Calinski–Harabasz index; at the same time the largest cluster is larger for the human – the human cloud has a more pronounced dense core. The distribution of cluster sizes is shown in Fig. 1.

Table 3: Wishart clustering, configuration SVD n=4, m=16

Metric	Human	Bot
Number of clusters	275	293
Noise fraction	72.0 %	72.5 %
Silhouette	0.359	0.366
Davies–Bouldin index	0.953	1.158
Calinski–Harabasz index	235	329
Largest cluster	1.82 %	1.64 %

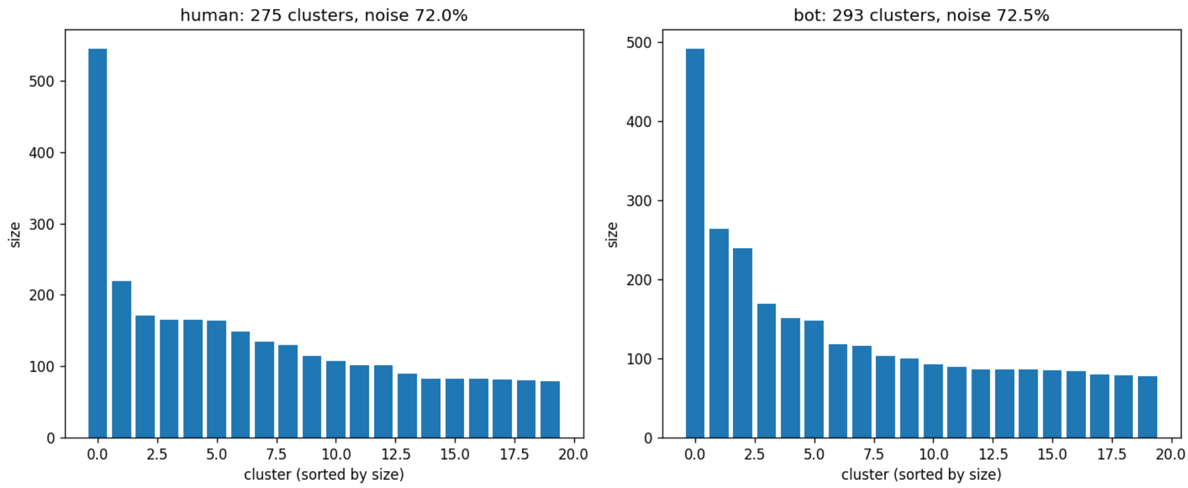

Figure 1: Distribution of cluster sizes (descending) for the human and bot clouds, configuration SVD n=4, m=16.

8.4 Assessing statistical significance

To make sure the differences are not an artifact of a particular subsample, a bootstrap estimate was carried out: $B = 40$ times a new independent subsample of 30,000 n-grams was drawn for the human and for the bot, and for each one the Wishart clustering was computed. Since the human and bot subsamples are independent, the *two-sample* Mann–Whitney test was applied; the confidence interval was built for the *mean* difference $\Delta = \overline{x}_B - \overline{x}_H$ via the standard error, and the effect size – Cohen's d.

The result (Table 4): **4 of the 5 metrics distinguish human and bot in a statistically significant way** ($p < 10^{-5}$, Cohen's effect size 1.2–2.3 – large). Only the silhouette metric is insignificant – this explains why its sign "floated" from configuration to configuration.

Table 4: Significance of the differences (bootstrap $B = 40$, SVD $n{=}4$, $m{=}16$). An asterisk marks the significant metrics ($p < 0.05$, the CI does not include 0)

Metric	Human	Bot	Δ	95 % CI Δ	p	d
Number of clusters	276.2 ± 8.3	289.1 ± 10.1	$+12.9$	$[8.9;\ 16.9]$	$5{\cdot}10^{-7}$	1.40^{*}
Noise fraction	0.709 ± 0.010	0.730 ± 0.009	$+0.021$	$[0.017;\ 0.026]$	$3{\cdot}10^{-12}$	2.27^{*}
Davies–Bouldin	1.001 ± 0.087	1.144 ± 0.138	$+0.143$	$[0.093;\ 0.194]$	$3{\cdot}10^{-6}$	1.25^{*}
Calinski–Harabasz	278.8 ± 38.4	348.9 ± 74.3	$+70.1$	$[44;\ 96]$	$2{\cdot}10^{-6}$	1.19^{*}
Silhouette	0.355 ± 0.012	0.361 ± 0.015	$+0.006$	$[-0.000;\ 0.012]$	0.081	0.41

9 Topological analysis

Persistent homology was applied to the human and bot clouds (configuration SVD $n{=}4$, $m{=}16$): for robustness the result was averaged over 5 subsamples of 2,000 points each, with homology computed up to H_1. The results – Table 5, the persistence diagrams – Fig. 2.

Table 5: Topological characteristics of the clouds (mean $\pm$ std. over 5 subsamples)

Quantity	Human	Bot
H_0, Betti number	1997 ± 2	1995 ± 3
H_1, Betti number (cycles)	1012 ± 21	1105 ± 28
H_1, total persistence	128.9	158.5
H_1, persistence entropy	6.41	6.42

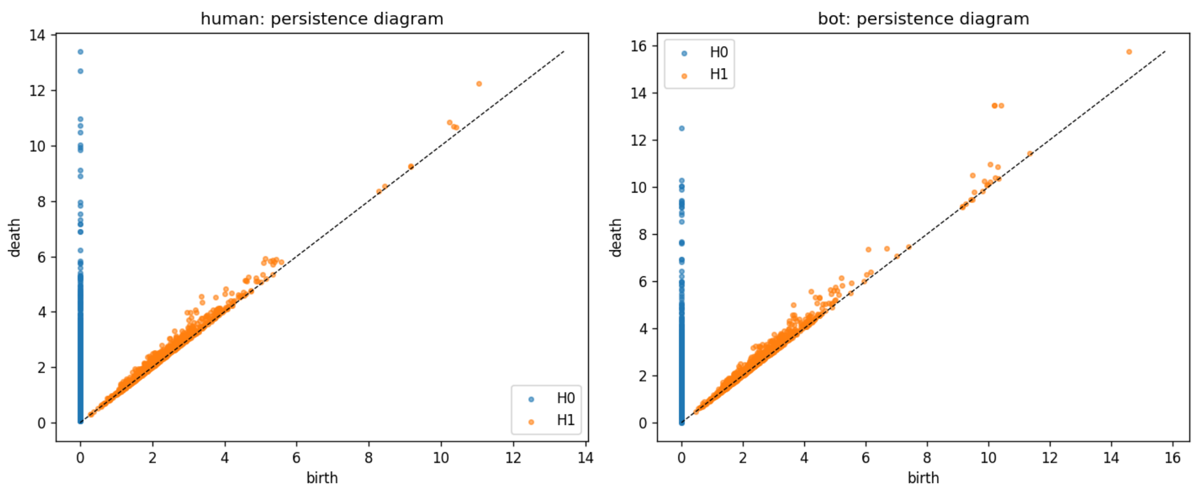

Figure 2: Persistence diagrams of the human and bot clouds (configuration SVD $n{=}4$, $m{=}16$).

The bot cloud contains about 93 more H_1 cycles and has a larger total persistence of the one-dimensional features. Taking into account the spread over subsamples, this difference is also significant (an approximate 95 % CI for the difference is $\approx [62;\ 124]$). The topological conclusion agrees with the clustering result: the bot cloud is *more complex* and *more fragmented*.

10 Discussion

Two independent methods – density-based clustering and topological analysis – give a consistent result. The human n-gram cloud is **more consolidated**: fewer clusters, a pronounced dense core (a larger largest cluster), fewer topological cycles. The bot cloud is **more fragmented and topologically more complex**: more clusters, a higher noise fraction, worse cluster separation, more H_1 cycles.

In substantive terms this can be interpreted as follows: a word-level LSTM, learning to predict the next word, reproduces *local* collocability well, but does not reproduce the *global* organization of the semantic space – a human text "contracts" toward a stable core of themes and constructions, whereas a generated one spreads out into many small, poorly separated clumps.

Limitations. The noise fraction in the clustering is high (about 72 %) – a consequence of the curse of dimensionality; the absolute values of the metrics should be treated with caution, whereas the *differences* between the corpora are stable and significant. The significance parameter h of the Wishart algorithm turned out to be uninformative in a high-dimensional space. The word2vec embeddings, under a naive truncation of components, are unusable – a separate adaptation is required.

Directions for development. It makes sense to: study the influence of the generation temperature and of the LSTM quality on distinguishability; compare with other generator architectures; apply feature normalization before clustering; extend the topological analysis to H_2 and to comparing diagrams by the Wasserstein metric.

11 Conclusion

In this work the full pipeline of the *Spot the bot* project was carried out for Maltese: a corpus of 36,452 texts was collected and preprocessed, an SVD dictionary was built ($93{,}343 \times 1024$), a word-level LSTM generator was trained (perplexity 98) and a bot corpus of comparable size was produced, n-gram clouds were formed, and Wishart clustering and topological analysis were carried out.

The main result: the human and generated corpora are *statistically significantly distinguishable* by the structure of the n-gram cloud. Four of the five clustering metrics distinguish them with $p < 10^{-5}$ and a large effect size; the topological analysis independently confirms this: the bot cloud contains more one-dimensional cycles. Thus, for the Maltese language, the project's original hypothesis is confirmed: a generated text differs from a human one at the level of the holistic geometric and topological structure.

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